

Assignment - 1

Subject: UCSH-502 (Computer Networks)

Q1) Describe the role of the Application Layer in the OSI model. Explain HTTP and FTP protocols in detail.

The Application Layer is the topmost layer of the OSI model. It provides a user interface and enables applications to communicate over the network. It serves as a bridge between the user and the underlying network services, offering protocols and services for file transfer, email, remote login, and more.

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The Application Layer in the OSI model is the link between end-user applications and network services. It provides the environment for software processes on different devices to communicate and access network resources efficiently.

A key role of this layer is **process-to-process communication**, enabling applications on one device to interact with those on another for tasks like messaging, file transfers, or web browsing.

It uses various **protocols** for specific purposes, such as **HTTP** for web access, **SMTP** for email, **FTP** for file transfers, **DNS** for domain resolution, and **SNMP** for network management.

By ensuring data is correctly formatted, transmitted, and interpreted, the Application Layer supports reliable, meaningful interaction between users across a network.

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Key Functions of the Application Layer:

- Provides network services directly to user applications such as email (SMTP), web browsing (HTTP), and file transfer (FTP).
- Facilitates resource sharing and remote file access.
- Handles data encoding, dialogue control, and application-specific functionalities.

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- **Process-to-Process Communication** – Enables direct interaction between software applications running on different hosts.

- **Network Services to Applications** – Provides services like file transfer, email, web access, and remote login.
- **Data Representation** – Ensures that data is presented in a format understandable to both sender and receiver.
- **Resource Sharing and Remote Access** – Allows applications to access and share resources over the network.
- **Protocol Support** – Uses various protocols (HTTP, FTP, SMTP, DNS, SNMP, etc.) to facilitate communication and specific services.
- **User Interface to the Network** – Acts as the gateway for users to interact with network services through applications.

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HTTP (HyperText Transfer Protocol):

- A protocol used for transferring hypertext documents (web pages) on the World Wide Web.
- Operates on a client-server model where the client (browser) sends an HTTP request and the server responds with the requested data.
- Uses methods like GET, POST, PUT, and DELETE.
- Typically operates over TCP port 80 (HTTP) and port 443 (HTTPS for secure communication).

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1. Hypertext Transfer Protocol (HTTP)

HTTP is the foundation of web communication, enabling browsers to fetch and display web pages from servers. It works on a **request-response model** over TCP (port 80) and is **stateless**, meaning each request is independent. Modern **HTTPS** secures data with SSL/TLS encryption.

Highlights:

- Uses **methods** like GET, POST, PUT, DELETE.
- Supports **multimedia content** transfer.
- Relies on cookies/sessions to maintain user state.

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FTP (File Transfer Protocol):

- A protocol used to transfer files between a client and a server.

- Supports two modes: Active and Passive.
- Uses two channels: Control channel (commands) and Data channel (file transfer).
- Operates over TCP ports 20 (data) and 21 (control).
- Provides features like authentication, directory listing, and file manipulation.

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2. File Transfer Protocol (FTP)

FTP is designed for transferring files between a client and server. It uses two TCP connections: **Control (Port 21)** for commands and **Data (Port 20)** for file transfer. Variants like **FTPS** and **SFTP** add encryption for secure transfer.

Highlights:

- Enables **upload, download, and file management**.
- Supports **user authentication** for access control.
- Works in **Active or Passive** modes for compatibility.

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Q2) What is DNS? Explain its purpose and working with an example.

DNS (Domain Name System) is a distributed database system that translates human-readable domain names (like `www.example.com`) into IP addresses (like `192.168.1.1`). This translation is crucial because computers use IP addresses to communicate, while humans prefer easy-to-remember names.

Purpose of DNS:

- Provides a hierarchical naming system for network resources.
- Allows users to access websites using domain names instead of numeric IP addresses.
- Supports email routing and other services using MX and other DNS records.

Working of DNS:

1. A user enters a URL (e.g., `www.example.com`) in the browser.
2. The browser checks its local cache; if not found, it queries the recursive resolver.
3. The resolver contacts the root DNS server to find the Top-Level Domain (TLD) server (`.com`).
4. The TLD server provides the authoritative DNS server for `example.com`.
5. The resolver queries the authoritative server, which responds with the IP address.
6. The browser uses the IP address to connect to the web server.

Example: Resolving www.google.com involves the browser querying the DNS hierarchy to obtain Google's IP address.

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Domain Name System (DNS)

The **Domain Name System (DNS)** is a fundamental part of the internet that translates human-readable domain names, such as www.google.com, into numerical IP addresses, like `142.250.182.14`, which computers use to identify and communicate with each other. Without DNS, users would have to remember complex IP addresses to access websites, which would be inconvenient and impractical.

Purpose of DNS

The primary purpose of DNS is to make internet navigation easier and more user-friendly. Humans find it easier to remember names than numbers, while computers require numerical IP addresses to establish connections. DNS acts as a “phonebook” of the internet, mapping domain names to IP addresses so that when a user types a web address into their browser, the correct website is retrieved quickly and efficiently. Additionally, DNS supports other services like email delivery, load balancing, and ensuring network scalability.

How DNS Works (with Example)

When you type www.example.com into your web browser, several steps occur behind the scenes:

1. **DNS Query Initiation:** The browser checks its local cache to see if it already knows the IP address. If not, it sends a DNS query to the local DNS resolver (usually provided by your Internet Service Provider).
2. **Recursive Resolution:** The resolver contacts a **root DNS server**, which responds with the address of a **Top-Level Domain (TLD) server** for `.com`.
3. **TLD Server Lookup:** The resolver queries the TLD server, which replies with the address of the **authoritative DNS server** for example.com.
4. **Authoritative Response:** The authoritative server provides the IP address associated with www.example.com.
5. **Website Access:** The resolver returns this IP address to your browser, which then uses it to establish a connection with the web server hosting the site.

For example, if you want to visit www.wikipedia.org, DNS ensures your browser finds the correct IP address of Wikipedia's server without you needing to know or type the actual IP address.

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Q3) Explain the major design issues of the Data Link Layer. Why are error detection and flow control important?

The Data Link Layer ensures reliable data transfer between directly connected nodes. It organizes bits into frames and handles transmission errors and flow control.

Major Design Issues:

- Framing: Dividing data into manageable units called frames for transmission.
- Error Control: Detecting and possibly correcting errors introduced during transmission.
- Flow Control: Ensuring that a fast sender does not overwhelm a slow receiver.
- Addressing: Identifying source and destination within a shared medium.
- Access Control: Managing access to a shared communication medium (e.g., CSMA/CD in Ethernet).

Importance of Error Detection:

- Transmission errors (noise, attenuation) can corrupt data.
- Error detection techniques like CRC, parity check ensure integrity.

Importance of Flow Control:

- Prevents buffer overflow at the receiver.
- Improves overall network efficiency and reliability.

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The **Data Link Layer** is crucial in ensuring reliable communication between directly connected nodes in a network. It sits just above the physical layer and deals with the transmission of data frames over a link. While its primary job is to ensure that data moves from one node to another without errors, several design issues make this possible:

Major Design Issues of the Data Link Layer:

1. Framing:

Data from the network layer is a continuous stream of bits, but the Data Link Layer must divide this stream into manageable chunks called **frames**. Proper framing ensures that the receiver can identify where each frame starts and ends. Techniques like character count, byte stuffing, or bit stuffing are often used.

2. Addressing:

In networks with multiple devices, frames need **addresses** (source and destination) to ensure they reach the intended recipient on a shared medium.

3. Error Detection and Handling:

Transmission over a physical medium is prone to errors caused by noise, signal attenuation, or interference. The Data Link Layer uses mechanisms like

- checksums, parity bits, and cyclic redundancy checks (CRC)** to detect errors in received frames.
4. **Flow Control:**
If a sender transmits data faster than the receiver can process it, data loss can occur. Flow control techniques, such as **stop-and-wait** or **sliding window protocols**, prevent overwhelming the receiver.
 5. **Access Control:**
In shared networks (like LANs), multiple devices may attempt to use the same communication medium simultaneously. The Data Link Layer uses **media access control (MAC)** techniques like CSMA/CD (Ethernet) or token passing to manage this.
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Importance of Error Detection and Flow Control:

Error Detection ensures data integrity. Without it, corrupted data might reach higher layers, leading to application failures, wrong outputs, or security risks. Detecting errors early allows retransmission and maintains reliable communication.

Flow Control prevents data loss due to buffer overflow at the receiver's end. It ensures smooth communication by matching the data rate of the sender with the receiver's ability to process it.

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Q4) Describe the Stop-and-Wait Flow Control Protocol with the help of a diagram.

Stop-and-Wait is a simple flow control protocol where the sender transmits one frame at a time and waits for an acknowledgment (ACK) before sending the next frame.

Working:

1. Sender sends a frame and starts a timer.
2. Receiver receives the frame, processes it, and sends back an ACK.
3. Upon receiving ACK, the sender sends the next frame.
4. If no ACK is received before the timer expires, the sender retransmits the frame.

Advantages:

- Simple to implement.
- Ensures reliable delivery.

Disadvantages:

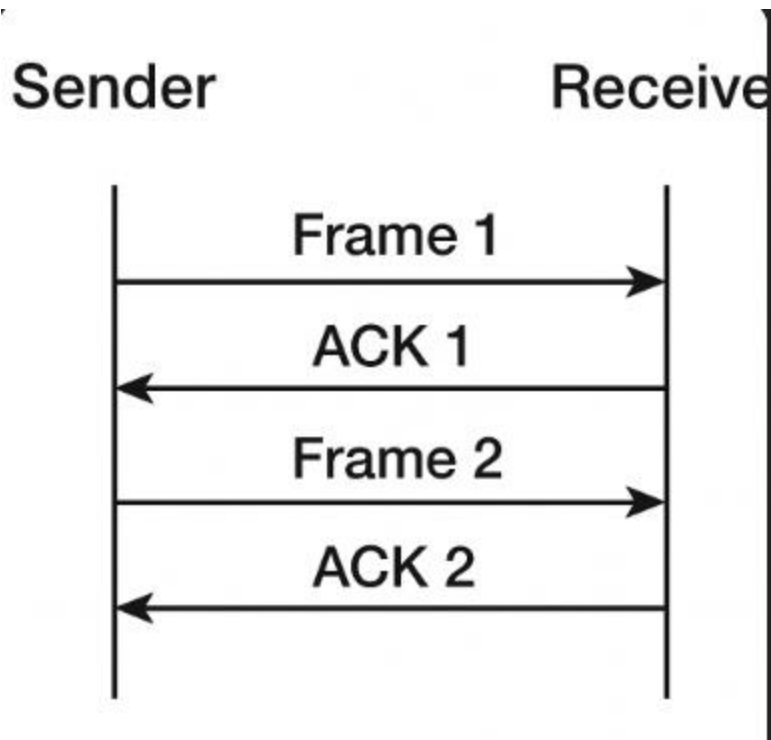
- Diagram (Description for insertion):
- ```
[Sender] ---> Frame1 ---> [Receiver]
[Sender] <--- ACK1 <--- [Receiver]
[Sender] ---> Frame2 ---> [Receiver]
```
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The **Stop-and-Wait Flow Control** protocol is one of the simplest methods used in data communication to ensure reliable delivery of data frames between a sender and a receiver. It operates on the principle that the sender transmits one frame at a time and waits for an acknowledgment (ACK) from the receiver before sending the next frame.

1. **Frame Transmission:** The sender transmits a single data frame to the receiver.
2. **Waiting for ACK:** After sending, the sender stops and waits until it receives an acknowledgment from the receiver.
3. **Acknowledgment Reception:** The receiver, upon receiving the frame, sends an acknowledgment back to the sender indicating successful receipt.
4. **Next Frame Transmission:** Once the sender receives the ACK, it sends the next frame. If the ACK is not received within a specified time (due to errors or loss), the sender retransmits the frame.

This method ensures that frames are delivered accurately and in order but can lead to inefficiency if the communication link has a high delay, as the sender remains idle while waiting.

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Sender Receiver
|-----Frame 1----->|
| |
|<-----ACK 1-----|
|-----Frame 2----->|
| |
|<-----ACK 2-----|
```



In this diagram, the sender sends **Frame 1**, then waits until **ACK 1** is received before sending **Frame 2**. This continues for subsequent frames.

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**Q5) Write short notes on any two of the following:**

**a) Error Detection and Correction Methods**

- Error Detection: Methods like Parity Check, Checksum, and Cyclic Redundancy Check (CRC) detect errors in transmitted data.
- Error Correction: Techniques like Hamming Code and Forward Error Correction (FEC) not only detect but also correct errors without retransmission.

**b) Sliding Window Protocols**

- Sliding Window Protocols allow multiple frames to be sent before needing an acknowledgment, improving efficiency.
- Windows define the range of sequence numbers allowed in transmission.
- Variants: Go-Back-N (retransmit from error onward) and Selective Repeat (retransmit only errored frames).

## Q6) Write notes on Wireless LAN.

Wireless LAN (WLAN) is a local area network that uses radio waves instead of cables for communication.

Features:

- Mobility: Allows users to connect without physical constraints.
- Flexibility: Easy to deploy and expand.
- Standards: IEEE 802.11 family (Wi-Fi).

Components:

- Access Points (AP): Act as central hubs.
- Wireless Clients: Laptops, smartphones, IoT devices.

Operation:

- WLANs use spread-spectrum technology (DSSS, OFDM) for reliable wireless transmission.
- Provide security features like WPA2 for data protection.

Applications:

- Office networks, public hotspots, home networking.

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### Wireless LAN (WLAN)

A **Wireless Local Area Network (WLAN)** is a type of local area network that uses radio waves instead of physical cables to connect devices within a limited area, such as homes, offices, schools, or campuses. It allows users to access and share network resources like the internet, printers, and files without the need for wired connections, providing flexibility and mobility.

#### Features of WLAN:

1. **Mobility** – Users can move freely within the network coverage area while maintaining connectivity.
2. **Ease of Installation** – No cabling is required, making setup and expansion simpler and more cost-effective.
3. **Scalability** – New devices can be added easily without additional wiring.
4. **Standards** – Most WLANs follow IEEE 802.11 standards, commonly known as Wi-Fi.

#### How WLAN Works:

A WLAN typically consists of **Access Points (APs)** and **wireless devices**. The AP connects to the wired network and acts as a bridge for wireless devices like laptops, smartphones, or IoT gadgets. Data is transmitted via radio frequency signals, and encryption methods like WPA2/WPA3 ensure security.

***Advantages of WLAN:***

- Greater flexibility and convenience compared to wired networks.
- Cost-effective for dynamic environments where wiring is impractical.
- Supports multiple devices simultaneously.

***Disadvantages of WLAN:***

- Susceptible to interference from other wireless devices.
- Security risks if not properly configured.
- Limited range compared to wired LANs.

**In short**, WLANs provide a convenient, efficient, and widely used solution for modern networking needs, enabling seamless connectivity without the restrictions of physical cables.

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